Microeconomics

Lecture 3 · Duality and the Slutsky equation

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What we know so far

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- Budgets and utility maximisation: what the consumer can choose.
- Marshallian demand, corners and Kuhn–Tucker conditions.
- Indirect utility and the envelope theorem.

What we will learn today

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- Duality: two ways to describe the same optimal choice.
- Shephard’s lemma and compensated price responses.
- Substitution and income effects: the Slutsky equation.

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Relative prices change: some goods become more expensive compared with others.

Purchasing power changes: the same income buys a different standard of living.

We need a way to separate these two effects.

The expenditure minimisation problem

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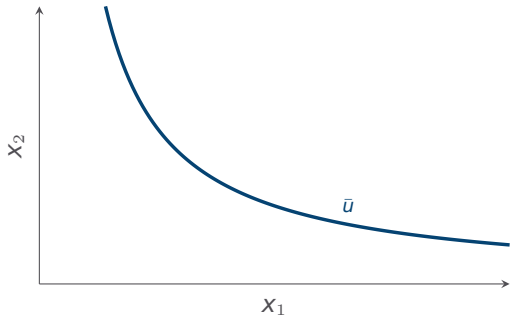
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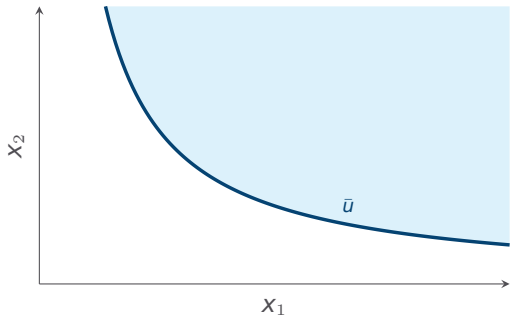
e: expenditure function. H: Hicksian demand correspondence; h if unique.

Reach a target indifference curve at minimum cost



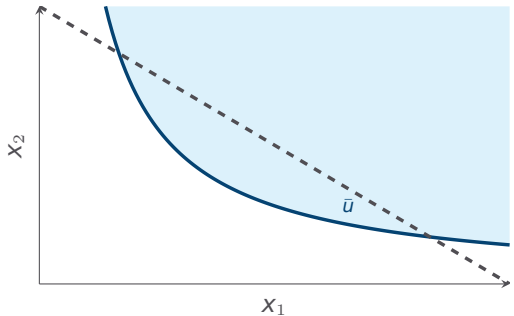
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Reach a target indifference curve at minimum cost



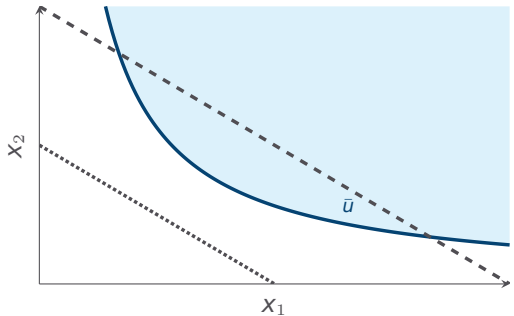
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The shaded region gives at least this utility.

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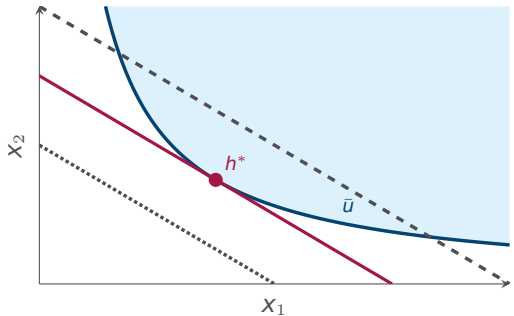
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The shaded region gives at least this utility.
A higher spending line reaches the target.

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Minimum spending: the cheapest line that reaches the target.

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- Add locally nonsatiated preferences for the duality results.
- Add uniqueness and smoothness only where derivatives require them.

Does a minimum exist?

Proposition. If u is continuous, $p \gg 0$ and $\bar{u}$ is attainable, then $H(p, \bar{u}) \neq \varnothing$.

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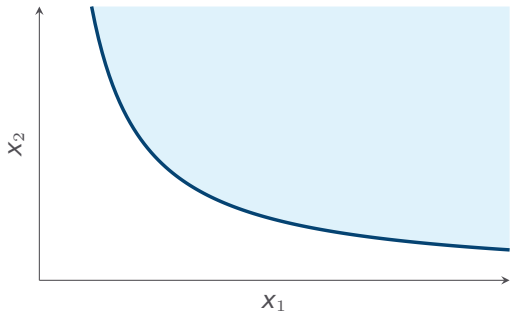
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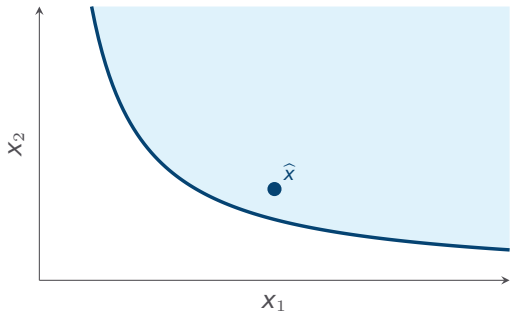
Find *one* feasible bundle and discard every bundle that costs more.

Cut away the expensive bundles



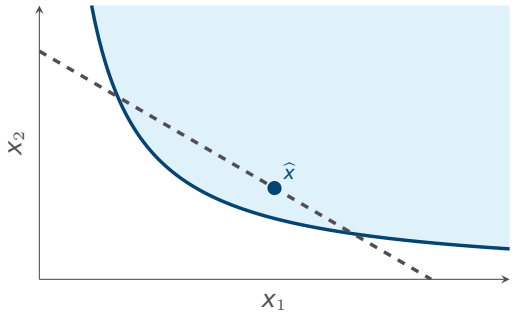
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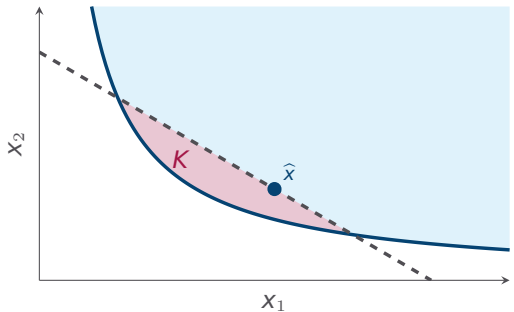
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$$K = \{x \geq 0 : u(x) \geq \bar{u}, \quad p \cdot x \leq p \cdot \widehat{x}\}.$$

Closed and bounded: **compact**.

Existence: cut away expensive bundles

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Every excluded feasible bundle costs more than $p \cdot \widehat{x}$, so this is a global minimum.

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Otherwise, continuity gives $t < 1$ close to 1 with $u(th) > \bar{u}$.

But $p \cdot (th) < p \cdot h$: a cheaper feasible bundle. Contradiction.

Change the currency, not the chosen bundle

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Hicksian demand has degree 0 in prices; expenditure has degree 1.

Convex preferences and Hicksian choices

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Strictly convex preferences imply a unique Hicksian choice.

Two distinct minimisers have utility exactly $\bar{u}$. Their mixture would give more than $\bar{u}$, violating no excess utility.

Buy a Cobb–Douglas standard of living

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Then evaluate your answer at $p = (4, 1)$ and $\bar{u} = 3$.

Cobb–Douglas: the dual solution

Minimise cost while attaining the target $u(x) \geq \bar{u} > 0$.

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Thus $e(p, \bar{u}) = 2\bar{u}\sqrt{p_1 p_2}$.

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At $p = (4, 1)$ and $\bar{u} = 3$: $h = (3/2, 6)$ and $e = 12$.

Perfect complements: find the cheapest bundle

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The direct utility has a kink, but expenditure is smooth in prices.

Perfect substitutes: which route is cheapest?

For $u(x) = x_1 + x_2$, $p \gg 0$ and $\bar{u} > 0$, solve

$$\min_{x \geq 0} \; p_1 x_1 + p_2 x_2 \qquad \text{subject to } x_1 + x_2 \geq \bar{u}.$$

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$p_1 = p_2$: $H = \{x \geq 0 : x_1 + x_2 = \bar{u}\}$.

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The equalities hold for *sets of choices*, including nonunique optima.

Proof: the cheapest way to buy the maximum

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If $e(p, \bar{u}) < m$, a minimiser h attains at least $\bar{u}$ at cost below m.

Then h is also utility-maximising with a slack budget, contradicting LNS and Walras. Thus $e(p, v(p, m)) = m$.

Proof: the most utility the minimum cost buys

Choose $h \in H(p, \bar{u})$ and set $m = e(p, \bar{u})$. No excess utility gives $u(h) = \bar{u}$.

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If an affordable z gave $u(z) > \bar{u}$, then z would also minimise expenditure for $\bar{u}$.

That violates no excess utility. Hence $v(p, e(p, \bar{u})) = \bar{u}$.

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Substitute $m = e(p, \bar{u})$ for the other equality.

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Verify that $h(p, v(p, m)) = x(p, m)$.

Checkpoint

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- We relabel utility as $\widetilde{u} = 2u + 10$. Which new target represents the old level $u = 7$?
- If utility maximisation has two optimal bundles, can the corresponding expenditure problem have a unique solution?

Shephard’s lemma

Fix an attainable target and positive prices. If the Hicksian minimiser is unique, expenditure is differentiable in prices and

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = h_i(p, \bar{u}).$$

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Interpretation: a small price increase costs the compensated consumer approximately $h_i \, dp_i$.

Recover quantities from cost

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Substitutes: $e = \bar{u}\min\{p_1, p_2\}$; differentiate only within a strict price regime.

The compensated law of demand

Fix $\bar{u}$. Compare two price vectors $p^1, p^2 \gg 0$ and choose

$$h^1 \in H(p^1, \bar{u}), \qquad h^2 \in H(p^2, \bar{u}).$$

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The utility target is the same in both choices; no differentiability is needed.

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h^1 is cheapest at p^1: $p^1 \cdot h^1 \leq p^1 \cdot h^2$.

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Add and rearrange:

$$p^1 \cdot h^1 + p^2 \cdot h^2 \leq p^1 \cdot h^2 + p^2 \cdot h^1 \quad \Rightarrow \quad (p^2 - p^1) \cdot (h^2 - h^1) \leq 0.$$

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$$p^1 \cdot h^1 + p^2 \cdot h^2 \leq p^1 \cdot h^2 + p^2 \cdot h^1 \quad \Rightarrow \quad (p^2 - p^1) \cdot (h^2 - h^1) \leq 0.$$

For differentiable unique demand, $\partial h_i / \partial p_i \leq 0$.

Symmetry of compensated price effects

NES

Suppose $e(\cdot, \bar{u})$ is C^2 locally. Then

$$\frac{\partial h_i}{\partial p_j}(p, \bar{u}) = \frac{\partial h_j}{\partial p_i}(p, \bar{u}).$$

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Mixed partials commute:

$$\frac{\partial h_i}{\partial p_j} = e_{p_i p_j} = e_{p_j p_i} = \frac{\partial h_j}{\partial p_i}.$$

Substitution has zero first-order cost

Fix $\bar{u}$ and assume differentiable Hicksian demand. For each price p_i,

$$\sum_j p_j \frac{\partial h_j}{\partial p_i} = 0.$$

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Differentiate $e(p, \bar{u}) = \sum_j p_j h_j(p, \bar{u})$:

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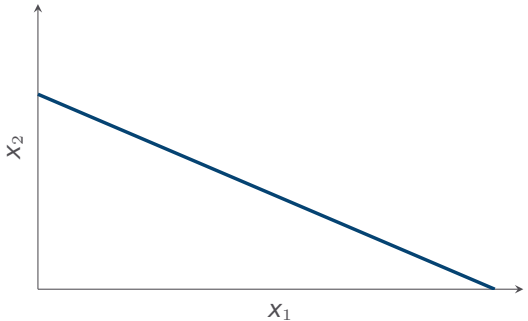
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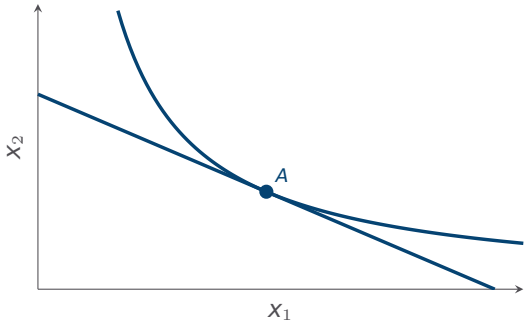
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By symmetry, also $\sum_j p_j \, \partial h_i / \partial p_j = 0$.

Substitution and income effects

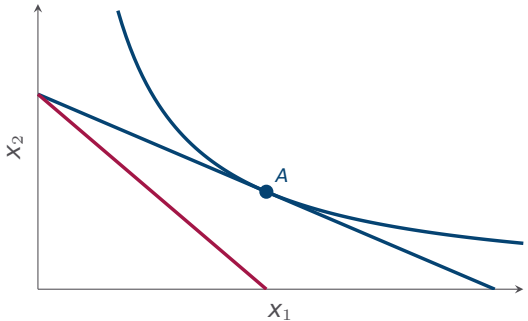


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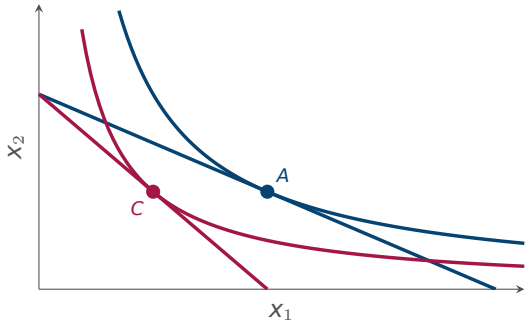
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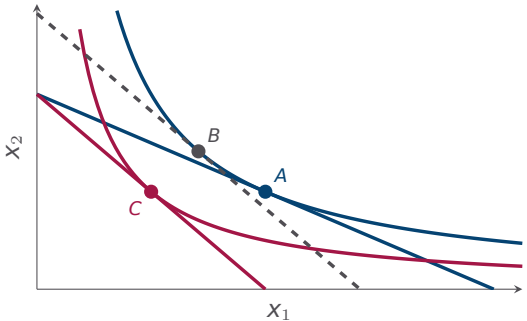
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Raise p_1, keeping income fixed.

Substitution and income effects



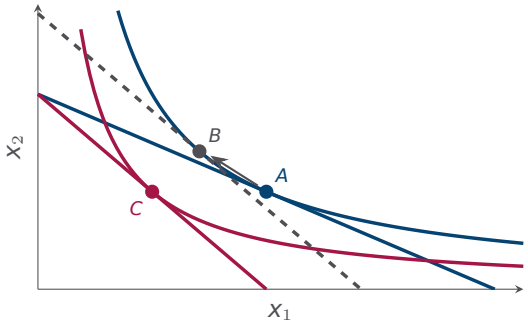
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Substitution and income effects



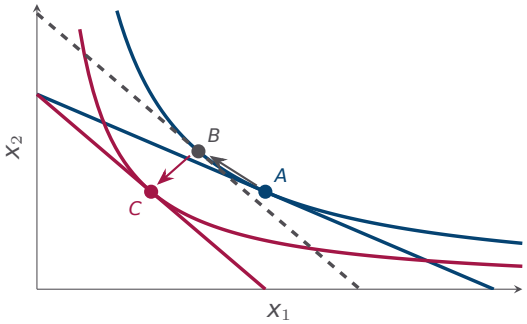
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Substitution and income effects



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$B \to C$: income effect.

The choices match. Their price derivatives do not.

$$h(p, \bar{u}) = x(p, e(p, \bar{u})).$$

Which two terms appear when we differentiate the right-hand side?

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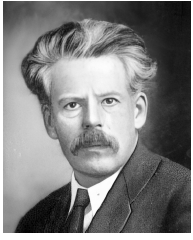
$$h(p, \bar{u}) = x(p, e(p, \bar{u})).$$

On the left, change prices while holding $\bar{u}$ fixed.

On the right, change prices *and adjust income* to $e(p, \bar{u})$.

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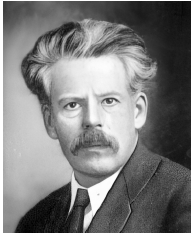
Eugen Slutsky and the consumer's budget



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1880–1948

Economist, mathematician and statistician.

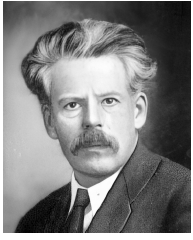
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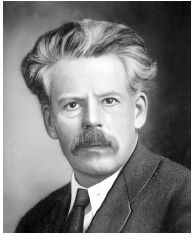
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"On the theory of the consumer's budget."
The price-effect decomposition we derive next bears his name.

Slutsky: the chain rule does the work

Assume unique demands with the derivatives below. Start from

$$h_i(p, \bar{u}) = x_i(p, e(p, \bar{u})).$$

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Shephard gives $e_{p_j} = h_j$. At $\bar{u} = v(p, m)$, $h_j = x_j$ and $e = m$.

The Slutsky equation

At $\bar{u} = v(p, m)$:

$$\underbrace{\frac{\partial x_i}{\partial p_j}}_{\text{total price effect}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{substitution effect}} - \underbrace{x_j \frac{\partial x_i}{\partial m}}_{\text{purchasing-power term}}.$$

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The quantity multiplying the income derivative is x_j, not x_i.

When must ordinary demand slope down?

For the own price:

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NES

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Giffen good: $\partial x_i / \partial p_i > 0$; the positive income effect dominates.

Checkpoint: normal and inferior goods

Assume positive prices, budget exhaustion and differentiable income responses.

- With two goods, both can be inferior at the same prices and income.

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- With three goods, two of them can be inferior.

Cobb–Douglas: split an own-price effect

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At $p = (1, 1)$, $m = 8$: total -4 = substitution -2 + income -2.

No total effect does not mean no substitution

Cobb–Douglas gives $x_1 = m/(2p_1)$, so $\partial x_1/\partial p_2 = 0$.

Does good 2 have no effect on the demand for good 1?

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The positive compensated substitution effect and negative income effect cancel exactly.

At $p = (1, 1)$, $m = 8$: $0 = 2 - 2$.

Perfect complements: all income, no substitution

For $u = \min\{x_1, x_2\}$:

$$x_1 = x_2 = \frac{m}{p_1 + p_2}, \qquad h_1 = h_2 = \bar{u}.$$

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All compensated price derivatives are zero.

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Ordinary demands fall together even though compensated quantities stay fixed.

Can a higher price make you buy more?

NES

At fixed tastes and income, a Giffen response means $\partial x_i / \partial p_i > 0$.

Can we construct a Giffen good with continuous, monotone, convex preferences?

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Yes. A smooth utility function is not needed for a smooth local demand.

Robert Giffen and the potato story



Robert Giffen

1837–1910

Economist and statistician.

Marshall’s example concerned bread.

- A poor household lives mostly on potatoes, with a little meat.

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- A poor household lives mostly on potatoes, with a little meat.
- Potatoes become more expensive. The same food budget now buys less.

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Economist and statistician.

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- A poor household lives mostly on potatoes, with a little meat.
- Potatoes become more expensive. The same food budget now buys less.
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Robert Giffen and the potato story



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- Potatoes become more expensive. The same food budget now buys less.
- It cuts meat and replaces those calories with potatoes, still the cheapest source of calories.
- Potato purchases can rise: the income effect outweighs substitution away from potatoes.

A subsistence kink generates Giffen demand

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The utility is continuous, strongly increasing and concave.

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$$\frac{\partial x_1}{\partial m} = -\frac{1}{2p_2 - p_1} < 0, \qquad \frac{\partial x_1}{\partial p_1} = \frac{10p_2 - m}{(2p_2 - p_1)^2} > 0.$$

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At the original point, $h_{1,p_1} = 0$ and Slutsky gives $3.125 = 0 - (2.5)(-1.25)$.

Checkpoint

- At one point, $x_1 = 2$, $x_{1,m} = -1$ and $h_{1,p_1} = -3$. Is good 1 Giffen?

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- If $x_{1,p_2} = 0$, must $h_{1,p_2} = 0$?
- Can raising only p_1 increase h_1 at a fixed utility target?
- Does a Giffen response prove that preferences are nonconvex?

Substitutes and complements: specify the experiment

For $i \neq j$, **net** substitution uses $\partial h_i / \partial p_j$: positive means net substitutes; negative means net complements.

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- A compensated own-price effect must be strictly negative.
- Gross cross-price effects must be symmetric.
- A Giffen good with positive consumption must be inferior at that point.

The consumer-theory square

Indirect utility

$v(p, m)$

Expenditure

$e(p, \bar{u})$

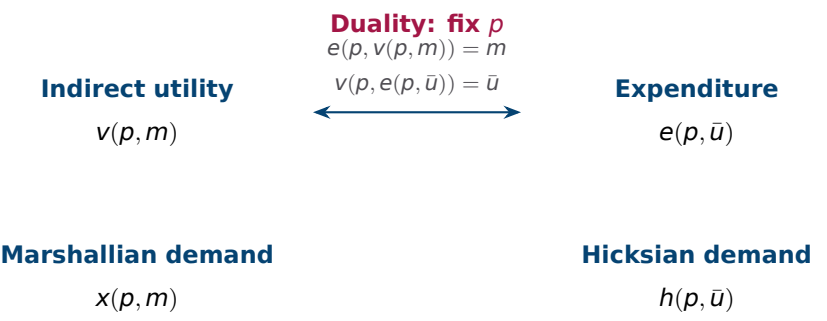
Marshallian demand

$x(p, m)$

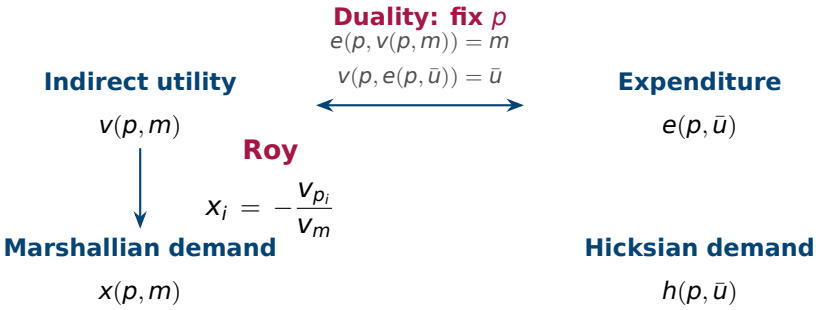
Hicksian demand

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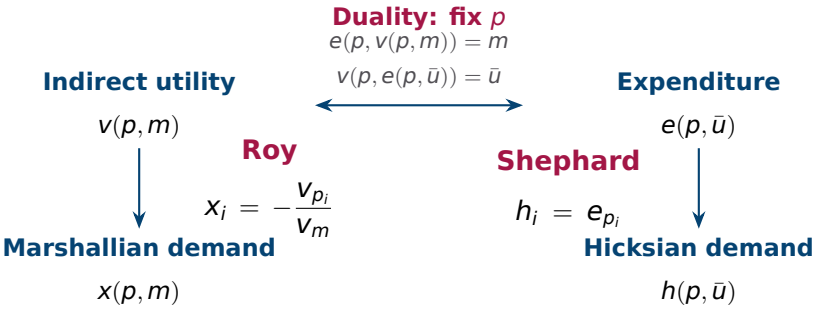
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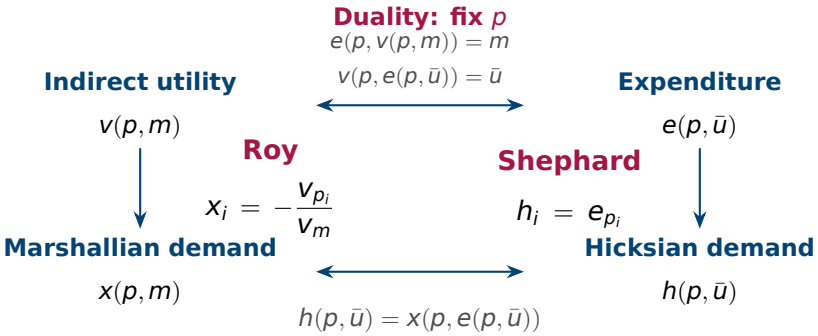
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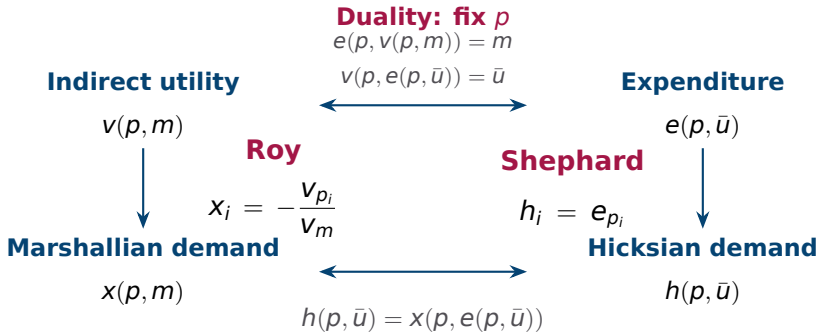
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Homework 1 covers Lectures 1–2 and is due **14 November 2026**.

Quasilinearity: income effects depend on the regime

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“No income effect” applies within the interior numeraire regime.

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Otherwise a minimiser for $\bar{u}'$ would also minimise cost for $\bar{u}$, contradicting no excess utility.

Expenditure is concave in prices

Proposition. For a fixed attainable target and $0 \le t \le 1$,

$$e(tp + (1-t)q, \bar{u}) \ge t\,e(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

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Why does minimising linear functions produce this direction of the inequality?

The same bundle must be purchased at the mixed prices. Reoptimising separately can only be cheaper.

Concavity: one bundle, two price vectors

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Therefore

$$\begin{aligned} e(tp + (1-t)q, \bar{u}) &= (tp + (1-t)q) \cdot z \\ &\geq t\, e(p, \bar{u}) + (1-t) e(q, \bar{u}). \end{aligned}$$

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With continuous LNS utility, e is continuous in positive prices and interior attainable targets.

Why the quantity appears in the derivative

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Divide by t and take limits; negative t gives the same derivative h_i.

What if demand is not unique?

For any $h \in H(p, \bar{u})$ and any $q \gg 0$,

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Use the demand correspondence at a kink; a single derivative cannot select the whole set.

The matrix behind compensated substitution

Assume $e(\cdot, \bar{u})$ is C^2 locally. Define

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In particular, $S_{ii} \leq 0$. Zero effects are allowed.

Why symmetry and the quadratic inequality?

By Shephard, $h_i = e_{p_i}$. Mixed partials of a C^2 function commute:

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Concavity of e implies concavity of g, hence

$$g''(0) = z^\top D^2_{pp} e(p, \bar{u}) z \leq 0.$$

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Relative prices drive compensated substitution; a change of currency does not.

A surprising restriction with only two goods

NES

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With three or more goods, negative net cross-effects are possible.

Construct compensated effects from ordinary demand

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Test: $S = S^\top$, $z^\top S z \leq 0$, $Sp = 0$, plus homogeneity and Walras.

Cobb–Douglas passes the matrix test

For $x_i = m/(2p_i)$,

$$S = \frac{m}{4}\begin{pmatrix} -1/p_1^2 & 1/(p_1p_2) \\ 1/(p_1p_2) & -1/p_2^2 \end{pmatrix}.$$

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At $p = (1,1)$, $m = 8$: $S = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$.

Could these choices come from our consumer model?

Suppose estimated demand is

$$x_1(p, m) = \frac{m p_1}{p_1^2 + p_2^2}, \qquad x_2(p, m) = \frac{m p_2}{p_1^2 + p_2^2}.$$

At $p = (1, 1)$ and $m = 2$, calculate the Slutsky matrix. Does the model survive?

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Start with $x = (1, 1)$ and $D_m x = (1/2, 1/2)^\top$.

A symmetric matrix can still fail

At $p = (1, 1)$, $m = 2$,

$$D_p x = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}.$$

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The demand violates compensated demand restrictions. It cannot be generated by the maintained consumer model.

Reserve: compensation and equivalent variation

For a price increase $p^1 \geq p^0$ at fixed income m, set $u^0 = v(p^0, m)$ and $u^1 = v(p^1, m)$.

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CV restores old utility at new prices; EV removes the equivalent income at old prices.

Reserve: how much compensation?

For $u = \sqrt{x_1 x_2}$, $p^0 = (1, 1)$, $p^1 = (2, 1)$ and $m = 8$:

$$u^0 = 4, \qquad u^1 = 2\sqrt{2}.$$

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These are exact finite-change measures; a derivative is only a local approximation.

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To first order they agree: $de = h(p^0, u^0) \cdot dp = x^0 \cdot dp$.

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This is the integrability problem. Converse theorems require additional domain and regularity conditions.

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Goods 1 and 2 are net complements:

$$\frac{\partial h_1}{\partial p_2} = -\frac{p_3}{(p_1 + p_2)^2} < 0.$$

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Those cost bounds make minimisers locally bounded. A convergent subsequence is target-feasible by continuity, giving $\liminf e(p^k, \bar{u}^k) \geq e(p, \bar{u})$.